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Vaccine Administration

By

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By the end of this session we should know

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- Types of vaccines
- Safety related to vaccination process
- Egyptian current compulsory vaccination schedule
- Additional non compulsory vaccines
- How to give the vaccine
- Practical schedule for **Primary** Vaccination

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Types of Vaccines

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Whole pathogen vaccine

1. Live vaccine — small pox
2. Live attenuated — oral polio, varicella
3. Killed vaccines — injectable polio, injectable flu, covid 19

Sub-unit Vaccines

1. **P**olysaccharide — 23 valent **P**PV, AC meningitis
2. **C**onjugate — P**C**V10, P**C**V13, MC**C**V

Sub-unit Vaccines

3. Toxoid — tetanus
4. Viral vector vaccines — Covid 19 — Astrazenica

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Nucleic Acid Vaccines

1. DNA Plasmid vaccines zika
2. Recombinant HBV
3. mRNA vaccines Covid 19 Pfizer

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Vaccination Safety

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The 7 Rights of Patient Safety

Right Patient

Right Time

Right Site

Right Route, Needle Length, Technique

Right Vaccine and/or Diluent

Right Dosage

Right Documentation

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Healthcare Personnel

1. Hand washing
2. Gloves
3. Sharp handling

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Egyptian Current Compulsory Vaccines

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1. Oral Polio Vaccine (Sabin)

- Live attenuated
- Strain 1 and 3
- Herd immunity
- 2 drops orally
- 0, 2, 4, 6, 9, 12, 18

2. Injectable Polio Vaccine (Salk)

- Killed vaccine
- No herd immunity
- Strains 1,2 and 3
- 0.5 ml IM left thigh
- 2, 4, 6

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3. Hepatitis B Vaccine

- Recombinant DNA
- 0 dose
- 0.5 ml IM right thigh
- 2, 4, 6 (part of the Penta vaccine)

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4. Diphtheria whole-cell Pertussis Tetanus

- Toxoids of D T, Polysaccharide capsule of P
- 2, 4, 6 (Part of the Penta vaccine)
- 18 0.5 ml IM right thigh

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5. Hemophilus Influenza type B

- Conjugate vaccine Tetanus Toxoid
- 2, 4, 6 (part of Penta vaccine)

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6. Bacillus Calmette–Guérin

- Live attenuated
- Within 1st 40 days of life
- Single 0.05 ml intradermal injection at the insertion of the left deltoid

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7. **M**easles **M**umps **R**ubella

- Live attenuated
- 12, 18
- 0.5 ml subcutaneous injection at the right arm

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Additional non compulsory vaccines

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Rota Virus

2 commercially available brands in Egypt

1. Rotarix (RV1)

   - Monovalent human strain
   - Protection against 9 strains
   - 2 dose schedule 2 month apart (4-10wks), orally
   - Can start as early as 6 weeks, must finish before 8 months

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Rota Virus

1. Rotarix (RV1)

   - Clear oral suspension in pre-filled oral applicator
   - 1.5 ml / dose
   - Store between 2-8 C, do not freeze



Rota Virus

2. Rotateq (RV5)

   - Pentavalent non-human strains
   - 3 dose schedule 2 months apart (4-10wks), orally
   - Can start as early as 6 weeks, must finish before 8 months

Rota Virus

2. Rotateq (RV5)

- Clear oral suspension in pre-filled squeezable tube
- 2ml / dose
- Store between 2-8 C, do not freeze



MeningoCoccal Conjugate Vaccine

2 commercially available Brands in Egypt

1. Nimenrix

   - A, C, W135, Y quadrivalent polysaccharide conjugate to Tetanus Toxoid
   - powder and solvent for solution for injection in pre-filled syringe
   - 0.5 ml IM after reconstitution

MeningoCoccal Conjugate Vaccine

1. Nimenrix

   - Administration as early as 6 weeks
   - Primary administration schedules and booster doses in different age groups
   - Administration in Immunodeficiency



MeningoCoccal Conjugate Vaccine

2. Menactra

- A, C, W135, Y quadrivalent polysaccharide conjugate to Diphtheria Toxoid
- powder and solvent for solution for injection in separate vials
- 0.5 ml IM after reconstitution

MeningoCoccal Conjugate Vaccine

2. Menactra

- Start after 9 month
- Primary administration schedules and booster doses in different age groups
- Administration in Immunodeficiency



Hepatitis A Virus

2 commercially available brands in Egypt

1. Havrix

- Suspension for injection in a adult dose 1 ml pre-filled syringe for IM injection
- Inactivated (killed) vaccine
- From 12 month to 18 years 2 doses 6-36m apart
- 0.5 ml per dose



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Hepatitis A Virus

2. Avaxim

- Suspension for injection in a pre-filled syringe for IM injection
- Inactivated (killed) vaccine
- From 12 month to 18 years 2 doses 6-36m apart
- 0.5 ml per dose



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Hepatitis A Virus

Both brands

- Interchangeable
- Store between 2-8 never freeze

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Pneumococcal Conjugate Vaccine

2 commercially available Brands in Egypt

1. Synflorix

- 10 strains (1, 4, 5, 6B, 7F, 9V, 14, 18C, 19F, 23F)
- Protection against 12 strain
- Administration as early as 6 weeks 2 months apart and booster between 12 to 18 m
- Primary administration schedules and booster doses in different age groups

Pneumococcal Conjugate Vaccine

1. Synflorix

- suspension for injection in pre-filled syringe
- conjugated to protein D carrier protien (derived from non-typeable Haemophilus influenzae)
- 0.5 ml for IM injection



Pneumococcal Conjugate Vaccine

2. Prevenar 13

- 13 strains (1, **3,** 4, 5, **6A,** 6B, 7F, 9V, 14, 18C, **19A,** 19F, 23F)
- Protection against 13 strain
- Administration as early as 6 weeks 2 months apart and booster between 12 to 18 m
- Primary administration schedules and booster doses in different age groups

Pneumococcal Conjugate Vaccine

2. Prevenar 13

- Powder and solvent in pre-filled syringe for IM injection
- Conjugated to CRM_{197} carrier protein (non toxic mutant of DT)
- 0.5 ml for IM injection



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Varicella Vaccine

2 commercially available Brands in Egypt

1. Varivax

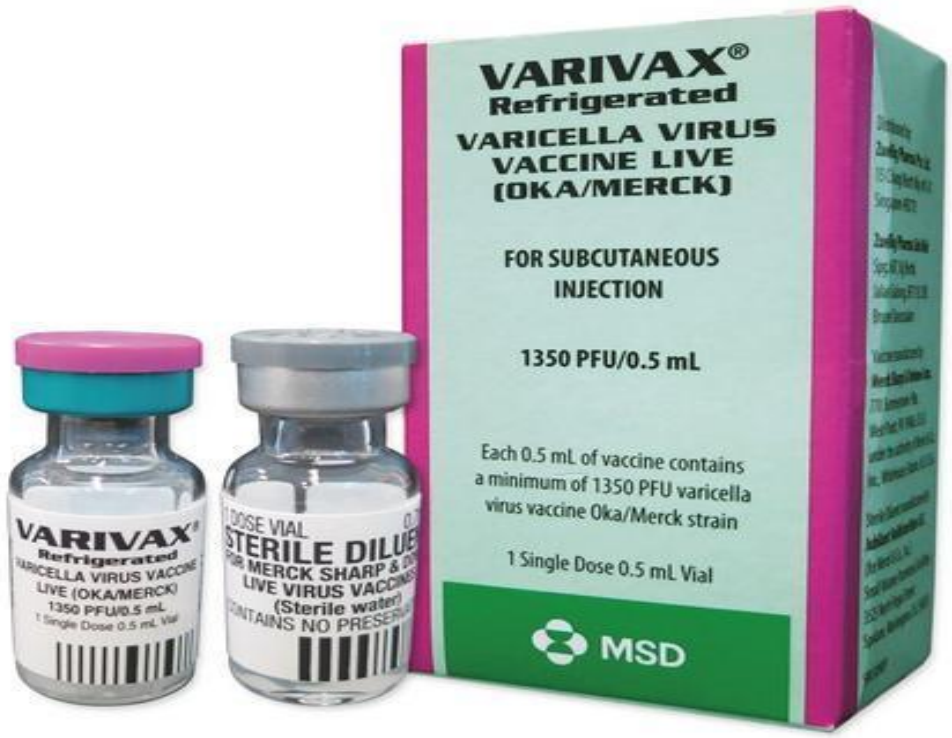
- Live attenuated
- powder and solvent for injection in separate Vials
- Start at age 12 m
- Primary administration schedules and booster doses in different age groups

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Varicella Vaccine

1. Varivax

   - 0.5 ml per dose after reconstitution
   - For IM or SC injection



Varicella Vaccine

2. Varilrix

- Live attenuated
- powder and solvent for injection in a pre-filled syringe
- Start at age 12 m
- Primary administration schedules and booster doses in different age groups

Varicella Vaccine

2. Varilrix

- 0.5 ml per dose after reconstitution
- For IM or SC injection



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Varicella Vaccine

- Interchangeable
- Store between 2-8 C , do not freeze

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Seasonal influenza Vaccine

Influvac tetra

- suspension for injection in pre-filled syringe
- surface antigen, inactivated
- For IM injection

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Seasonal influenza Vaccine

Influvac tetra

- Start as early as 6 month
- Administration schedules in different age groups
- 0.5 ml per dose



Human Papilloma Virus Vaccine

1. Gardasil

    - Suspension for injection in single dose vial.
    - Tetravalent (serotypes 6, 11, 16, 18)
    - Recombinant DNA

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Human Papilloma Virus Vaccine

1. Gardasil
    - Start as early as 9 years
    - For IM injection
    - 0.5 ml per dose



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Hexaxim

1. Diphtheria Toxoid
2. Tetanus Toxoid
3. Bordetella pertussis antigens
    - Pertussis Toxoid
    - Filamentous Haemagglutinin
4. Poliovirus (Inactivated)
    - Type 1 Type 2 Type 3
5. Hepatitis B surface antigen
6. Haemophilus influenzae type b (polysaccharide conjugated to Tetanus protein)

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Hexaxim

- suspension for injection in pre-filled syringe
- 0.5 ml Per dose
- For IM injection



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Vaccine Administration

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Remember

- Review vaccination history
- Assess needed vaccinations
- Screen for precautions / Contra indications
- Educate your patient
- Prepare, administer the vaccine/s
- Document the vaccination

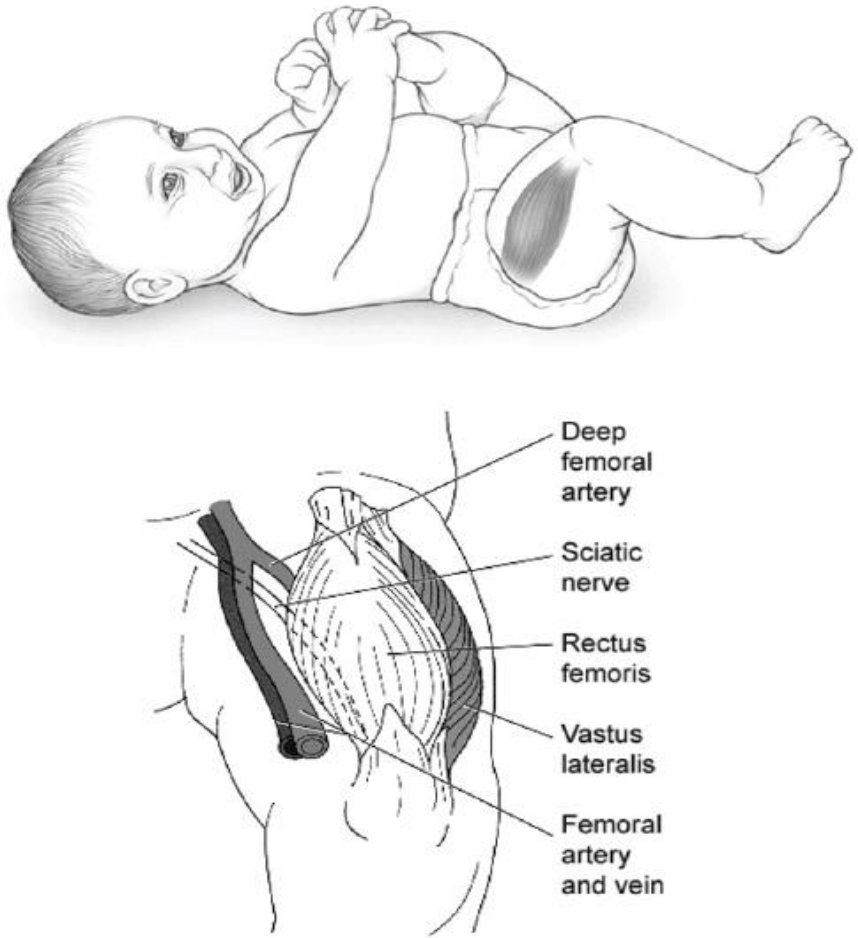
Remember

- U can give any number of vaccines in same session.
- Live vaccines are either given at same session or 28 day apart.
- Killed vaccines do not need to be separated by any time interval.
- When catching up Use the minimal interval and largest possible number of vaccines.

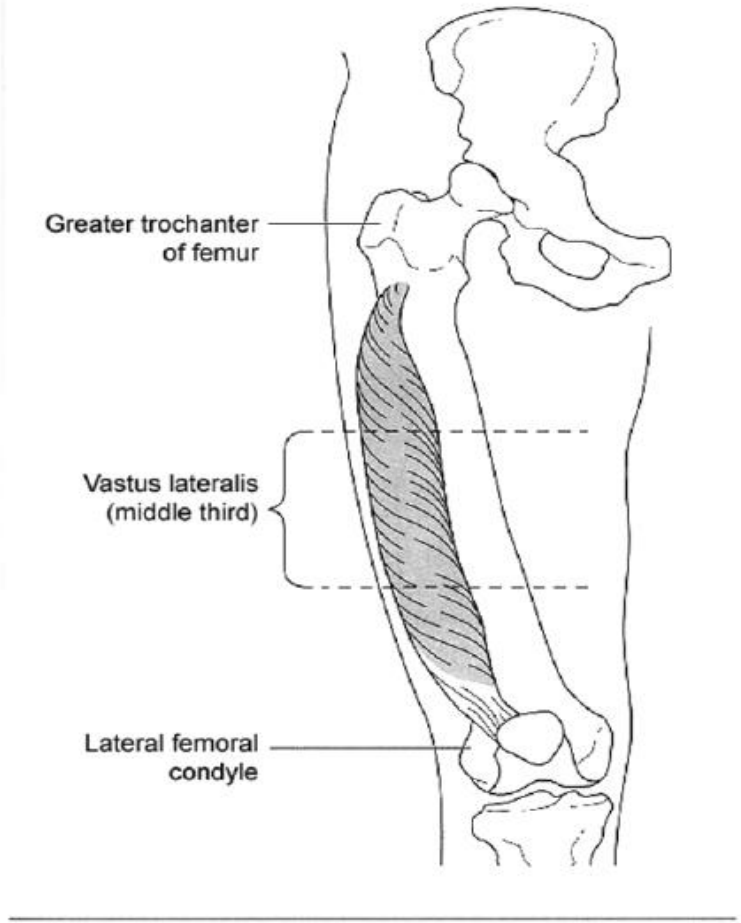
Intramuscular Injections	Site
• Preterm, neonates and infants	Anterolateral thigh
• Toddlers & older children	Deltoid or Anterolateral thigh
• Adolescents & adults	Deltoid

Subcutaneous Injections	Site
Infants	thigh
>12 months	Outer triceps

Intradermal Injections	Site
All age	Left deltoid

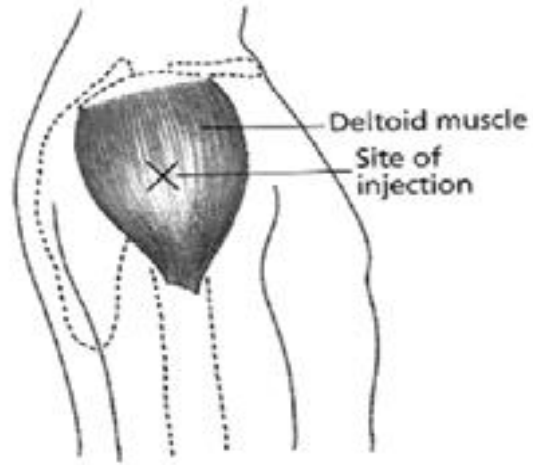


The vastus lateralis muscle of the upper thigh used for intramuscular injections.



The vastus lateralis site of the right thigh, used for an intramuscular injection.


Lynne Larson, www.biovisuals.com


Deltoid muscle
Site of injection

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Practical schedule for Primary vaccination

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Age	Vaccine/s	
1st day	OPV* , HBV*	
Before 40 days	BCG*	
2, 4, 6 month	OPV and Penta* Hexaxim	
3, 5, 7 month	Rota and PCV	
9 month	OPV*	
10 month	MCCV	
12 month	OPV* and MMR*	
13 month	MCCV	
15 month	HAV and varicella	
18 month	OPV*, MMR*, DwPT* PCV	
21 month	HAV	
4 year	varicella	
	* Vaccines given at MOH facilities	

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Any Questions??

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Thank you!

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References

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- http://www.emro.who.int/egy/programmes/expanded-programme-on-immunization.html